

Topic: Grid and Flex in Tailwind

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Tailwind CSS

Recap:

1. Colors using classes in Tailwind

`text-{color}-{shade}`

`text-red-500`

Range for shade - 100 to 900

100 - lightest shade

500 - normal color

900 - darkest shade

`bg-{color}-{shade}`

`bg-blue-300`

`bg-gradient-to-{direction} from-{color} via-{color} to-{color}`

`bg-[write custom css identifiers for multi color where colors > 3]`

2. Borders

`border-{width}` - adds a border

`border-{color}-{shade}` - to color the border

to add the color ensure border width is set using `border-{width}`

3. Typography

`text-{size}` -> xs, sm, md (default), lg, xl - 9xl

`text-[custom size]`

`font-{weight}` -> thin, light, normal, bold, extrabold

`leading-{height}` -> none, normal, relaxed

4. Spacing and Sizing

Width

`w-{unit}` -> unit is used to calculate the width using $0.25\text{rem} * \text{unit}$

`w-[custom width]`

Height

`h-{unit}`

`h-[custom-height]` -> same calculation performed

Margin

`m-{unit}`

`m-[custom]`

`mt / mr / mb / ml`

`mt + mb -> my`

`ml + mr -> mx`

Padding

$p\{-unit\}$
 $p[custom]$
 $p_t / p_r / p_b / p_l$
 $p_t + p_b \rightarrow p_y$
 $p_l + p_r \rightarrow p_x$

Rounded - to add round corners or border radius

Images

object-cover, object-contain, object-fill
to occupy the actual area in the element of the defined space

Flex - to align the contents of an element in 1D space

flex-direction to control the direction the content flow

like row or column

align the items in the center using

justify-content $\rightarrow$ main axis

align-items $\rightarrow$ cross axis

flex-direction is set to row by default

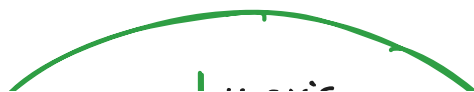
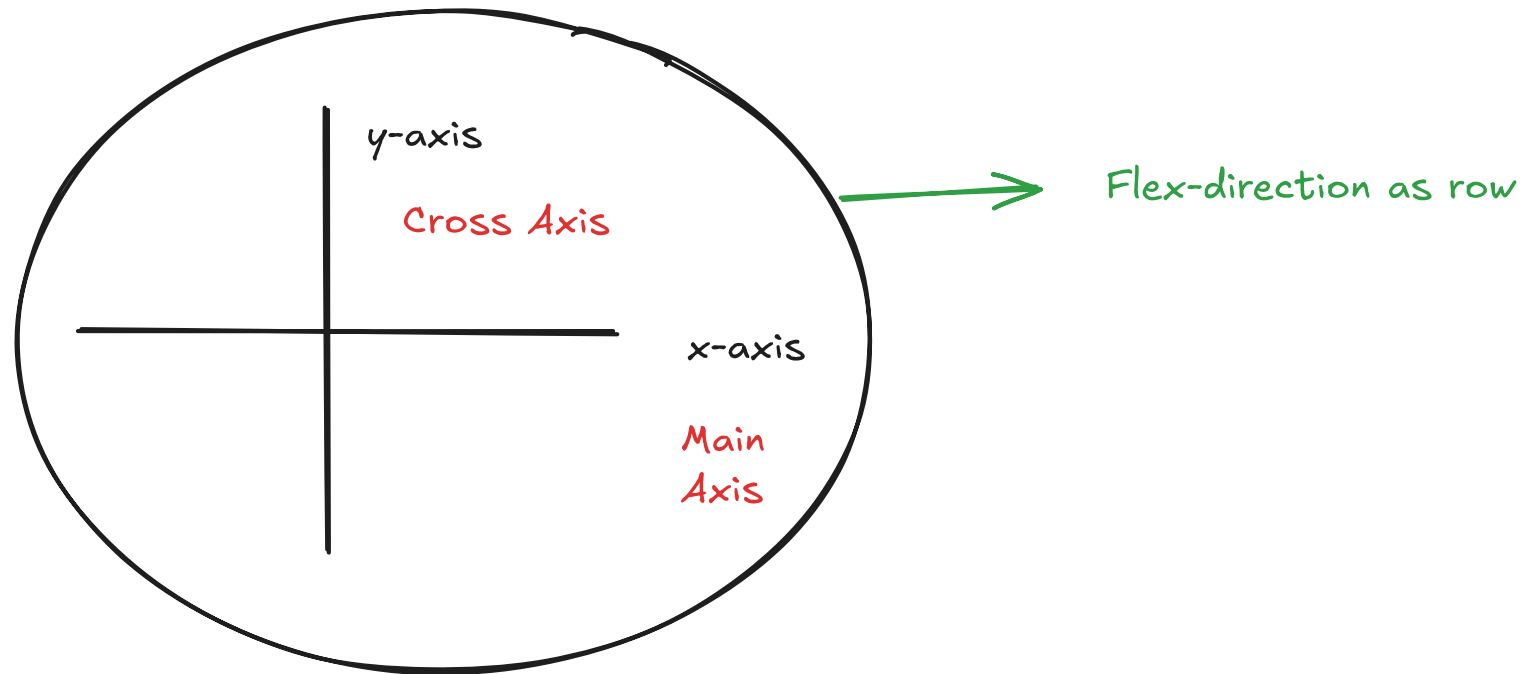
flex-direction: column

in tailwind we have two classes - flex-col

to enable flex in tailwind use "flex" class first
to define the direction use "flex-{direction}"

- row
- col
- row-reverse
- col-reverse

Align the content on main axis





`justify-content: center / flex-start / flex-end / space-between / space-around / space-evenly`

in the tailwind

`justify-{style property value}`

where value can start / end / between / around / evenly / center

to align items across cross axis in the tailwind

`items-{style property value}`

where value can be start / end / stretch / baseline / center

to align the content we can use

`content-{property value}`

where value is start / end / center / between / around / evenly

to add default width to flex item

`basis-{property value}`

`basis-auto`

basis-xs / xm / md / xl to 7xl / 3xs to xs
basis-[custom value]

flex grow property

grow -> it defaults to flex-grow as 1

grow-

flex: 1

flex-basis: occupy the minimum width

flex-grow: grow in a ratio reference of 1 to other siblings

flex-shrink: shrink in a ratio reference of 1 to other siblings

flex-[1_1_100px]

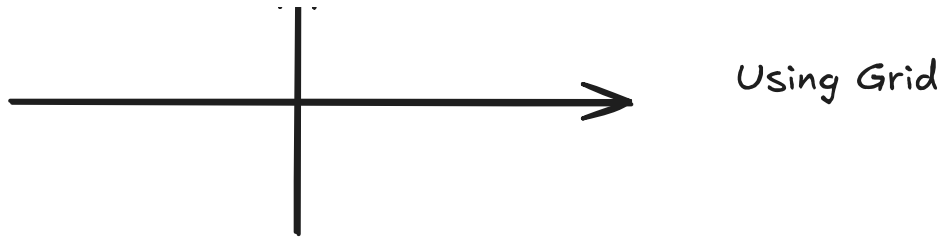
flex-grow: 1 -> it will grow in a ratio of 1:1 to other siblings, if no grow property is applied to sibling it will occupy the remaining space

flex-shrink: 1 -> it will shrink in a ratio of 1:1 to other siblings, if no shrink property is applied to sibling it will not occupy the remaining space.

flex-basis: 100px -> it provides a default width to the flex item

Grid - it aligns the items in 2D.





Grid Container - the element that has the display property of grid.

Grid Items - direct children of grid container are grid items

Grid Lines - the invisible lines that divide the grid into rows and columns

Grid Tracks - the space between two grid lines

Grid Cells - unit formed by the intersection of rows and columns

Grid Area - Area occupied by a grid item

In CSS

using `display: grid`, we can create a grid structure.

```
.grid {  
    display: grid; // to create grid  
    grid-template-columns: repeat(2, 1fr); // to define number of columns with size  
    grid-template-rows: 100px; // no of rows with the height  
    gap: 20px; // the track width  
}
```

`grid-template-columns` - define the number of columns for grid

`grid-template-rows` - define the number of rows in grid

`gap` - shorthand for `row-gap` and `column-gap`

`gap: 20px`

`row-gap: 20px`

column-gap: 20px

row-gap: defines the grid track size between rows

column-gap: defined the grid track size between columns

grid-column: it will tell the grid item how much cells should it occupy across columns

grid-row: it will tell the grid item how much cells should it occupy across rows

to define a grid area:

use

grid-template-areas:

Using Grid in Tailwind Format

to use grid set the class as "grid"

to define columns - grid-cols-{number}

to define rows - grid-rows-{number}

justify content and align content properties work in similar manner like flexbox

also align-items works in same manner as of flexbox

for grid items

1. col-span-{number} - for number of cells occupied across columns

2. row-span-{number} - for number of cells occupied across rows.